

Neurological Diseases

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In this course, we will first go over the specifics of the neurological examination in small mammals as some aspects are different than dogs and cats, then touch on the neurologic diagnostic work up, and finally discuss common and important neurological diseases of small mammals by species.

All in all, **neurologic diseases commonly affect rabbits** whereas neurologic signs are less commonly reported in ferrets and rodents and often appear to be a manifestation of systemic illness rather than a primary neurologic disease.

NEUROLOGICAL EXAMINATION

The neurological examination in small mammals follows the exact same steps as for any other mammal and is very similar to the dog and cat's exam. There are however a **few differences in its interpretation**, mainly due to the stoic nature of many of these species, which are, for the most part, prey species. We are not going to go into great details in the neurological examination of mammals, please refer to your neurology lectures. We will mainly summarize the neurological examination while highlighting specificities of small exotic mammals. We'll also **focus on the rabbit** in this section as many differences are found in this species and it is the most commonly seen small mammal here.

The primary aim of a neurologic examination is to determine whether a neurologic problem exists and, if so, to determine its anatomical location. When considered together with the patient's history and findings of physical examination, neuroanatomical localization allows the formulation of a list of possible differential diagnoses, which then further determines the diagnostic tests that can be performed to reach a definitive diagnosis.

The first part of the examination is the **observation** of the animal: mentation, gait (paresis, paralysis, ataxia,

circling, lameness), posture (head, trunk, limbs). A decrease level of consciousness may manifest as depression or stupor. Hamsters might be sleeping during the day and this should be taken into consideration. Also, some hamster species can go into torpor or hibernation.



For instance, see this [rabbit](#) affected by *Baylisascaris* infection, which is stuporous, cannot stand and is falling to its sides. Alteration of consciousness typically involve lesions in the brainstem, cerebrum, or metabolic issues and systemic illnesses. Posture and gait are affected by multiple CNS pathways and spinal reflexes. Head tilt and circling are commonly seen in rabbits and are typically indicative of central or peripheral vestibular dysfunction.

Something worth mentioning about these sneaky ferrets is that **ferrets are often ataxic in their back end** with a wide range of systemic diseases and these signs may be non-specific.

The next part is to assess **postural reactions**. Those include paw replacement, hopping, wheelbarrow, hemiwalking, and placing. But here's the catch: **most postural reactions are inconsistent in rabbits and rodents**. For instance, conscious proprioception by paw replacement is inconsistent in rabbits and chinchillas, except for the thoracic limbs in chinchillas and

guinea pigs. Hopping, wheelbarrow and hemiwalking are consistent in both rabbits and chinchillas, except wheelbarrow for the pelvic limbs in rabbits or hopping in the pelvic limb in guinea pigs, which are inconsistent. Placing reactions are inconsistent in the rabbit and the chinchilla, except in the rabbit in the pelvic limbs, in which they are completely absent and in the chinchilla in the thoracic limb, which are consistent.



The next step is evaluating **spinal reflexes**. These, as in other animals, include patellar, withdrawal, perineal, and cutaneous trunci reflexes. Most reflexes can be elicited in rabbits, ferrets, and rodents. **The withdrawal reflex is reliable.** The patellar reflex appears frequently as an exaggerated response followed by clonus or tremor in the rabbit. The perineal reflex is inconsistent and has a slow response in rabbits and absent in guinea pigs and chinchillas. The cutaneous trunci reflex is also inconsistent in rabbits and chinchillas and is difficult to interpret and almost absent in guinea pigs.



When that's done, it's the **cranial nerves** turn. Assessing cranial nerves first include assessing eye positions and reflexes such as menace response (II and VII), [dazzle](#) reflex (II, VII), pupillary light reflex (direct and consensual), II, III), oculoccephalic reflex (III, IV, VI, VIII), corneal reflex (V, VI, VII), palpebral reflex (V, VII). **Rabbits and rodents could not care less about responding to menace, so it is pretty much negative in normal rabbits, guinea pigs, and chinchillas.** In rabbits, direct PLR are slow and weak, while indirect PLR is rarely observed (only 10% of the optic nerve fibers decussate), and when present is weak and incomplete. In guinea pigs, the indirect PLR is inconsistent and in chinchillas, direct PLR is difficult to observe as it happens very quickly. All other reflexes are interpreted the same way as in other mammals.



Other tests focused on cranial nerves include jaw tone (V), nasal/ear sensation (V) and movement (VII), facial symmetry (VII), auditory stimulus (VIII), swallowing

(IX, X, XI) and tongue movement (XII). Trigeminal nerve (CN V) motor function by examining jaw tone is not considered appropriate in rabbits, due to the anatomy of these little animals which resent manipulation of the jaw, however, normal chewing behavior will presumably indicate normal motor functioning. For sensation focus on **nasal septum** (V sensory maxillary branch, see picture below), palpebral (V sensory medial cantus tests ophthalmic branch, lateral canthus tests maxillary branch) and outer ear canal (V sensory). **Ear movement inconsistent especially in lopped rabbits, and facial symmetry is important to evaluate (VII motor)** (see later). Detectable response to auditory stimulus is inconsistent in rabbits.

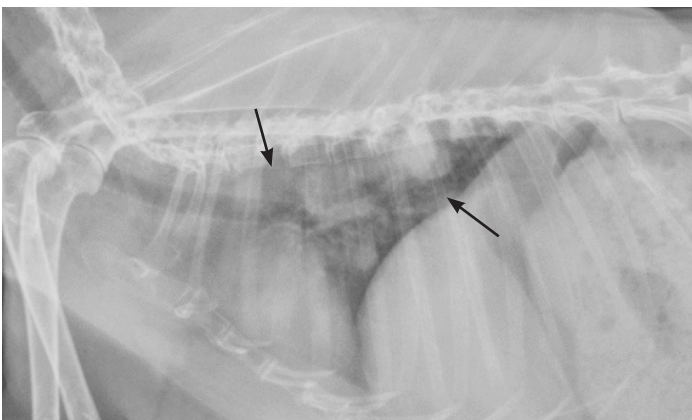


Pulling of the tongue to evaluate tonus and hence functioning of the hypoglossal nerve (CN XII) poses a risk of traumatizing the tongue; but can be observed when doing a dental examination with a speculum.

DIAGNOSTIC PLAN

Diagnostic Tests

The diagnosis workup of a neurological disease starts by performing routine tests such as blood work (CBC, biochemistry) and standard diagnostic imaging such as xrays of the skull and spine (see xray of a rabbit below as an example).

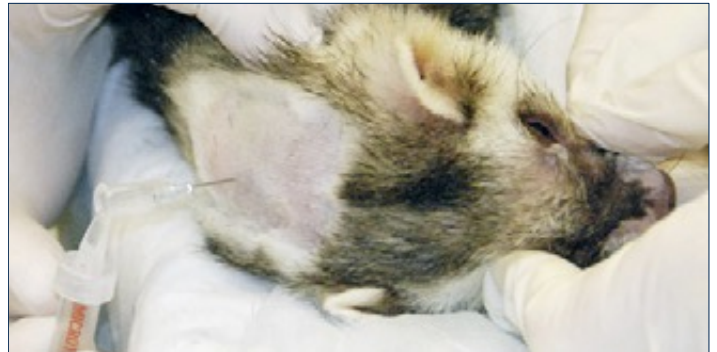


Specific infectious diseases may also be detected by

molecular testing or serology.

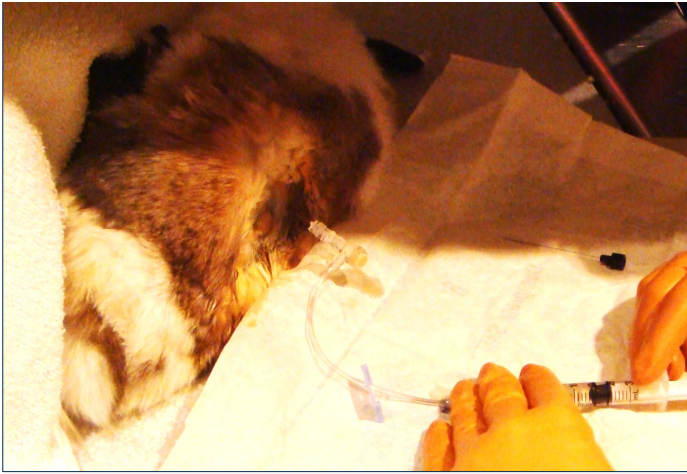
To go beyond that, specific diagnostic tests that may be performed for neurological diseases include CSF tap, myelogram, advanced imaging (CT, MRI), and electrodiagnostics (EMG).

CSF taps are definitely a bit trickier in ferrets and rabbits than in dogs and cats, but the technique is similar. It is **most commonly performed from the cisterna magna**. Only a small sample can be collected and typically not enough to do many tests, so a cytology or fluid analysis would be the only thing that could be done.



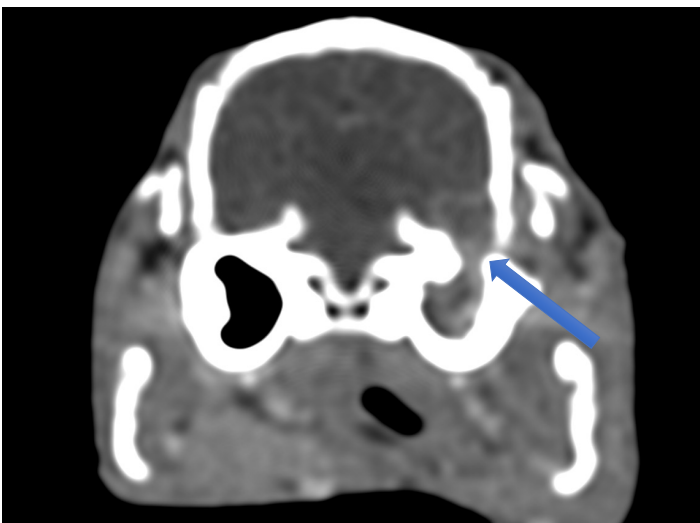
Myelograms are less commonly done now that CT imaging has become more accessible. The contrast medium can be injected into the ventral subarachnoid space at the level of the caudal lumbar vertebrae or into the cisternal space; however, death of rabbits after injection into the cisterna magna have been reported. The site routinely recommended for infusion of contrast medium during myelography is L5-L6. This site is recommended because it does not appear to result in injury to the rabbit spinal cord at that level; however; using a more caudal location (L6-L7 or L7-S1) may be preferred to minimize the effects on the spinal cord if there was a concern. The most common complications associated with myelography include seizures and the temporary or permanent exacerbation of clinical signs.

Other tests such as EEG, auditory or visual evoked potentials are rarely done in small mammals. Of special note is that most ferrets with a Panda or “blaze” pattern are deaf.

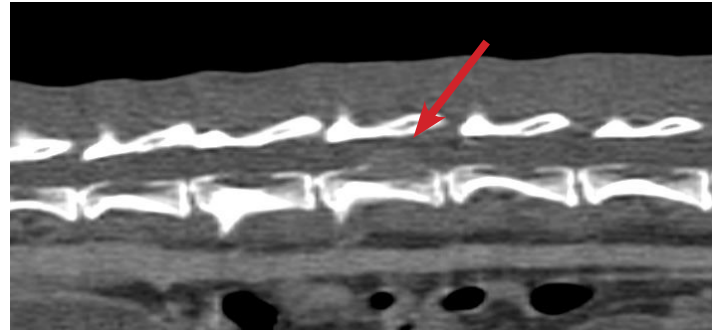


Rabbit: extradural spinal cord compression at T10 secondary to a vertebral lamina fracture

CT-scan is commonly used as an advanced imaging tool in small mammals as the scanning time is very short, animals don't have to be fully anesthetized, and the cost is much less than MRI. However, the soft tissue definition is not as good as MRI so it will only pick up obvious and significant lesions in the brain and spinal cord, but is very good for vertebral diseases and middle ear disease. Radiographs have a poor sensitivity to detect middle ear disease in rabbits.

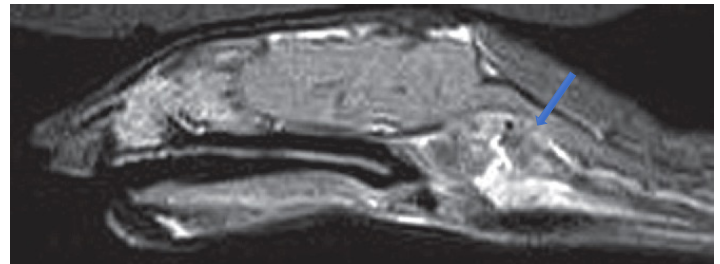


Middle and internal ear disease in a rabbit, extending into the brain, cross-sectional view

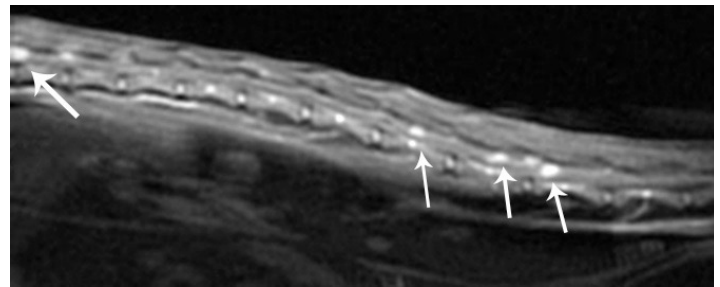


Mass in the spinal canal with extradural compression in a rabbit, sagittal view.

But the ultimate CNS imaging test is definitely MRI, which can use a wide array of different sequences to image the brain and spinal cord. It is not often used because of the cost, long time of scanning, the difficulties in monitoring the animals in the machine, and the coldness of the MRI room.



T1 sagittal view of a ferret, mass compressing spinal cord ventrally and extending into the bones of the skull



T1 sagittal view of another ferret post contrast administration. Multifocal hyperintense lesions in the spine, the ferret was diagnosed with spinal blastomycosis.

Differential Diagnosis

In order to be thorough, it is good practice to use the mnemonic VITAMIND:

- V** vascular
- I** Inflammatory/Infectious
- T** Toxic/Trauma
- A** Anomalies
- M** Metabolic
- I** Idiopathic/Iatrogenic
- N** Nutritional/Neoplastic
- D** Degenerative

It is especially useful for complex cases not to have tunnel vision.

RABBIT

We are not going to cover every single neurological disease in rabbits, just the most important or unique. Rabbits are also susceptible to a wide range of infections and neoplasias that won't be covered here. There are also some other non-infectious issues such as toxemia of pregnancy, **heat stroke**, or the **idiopathic epilepsy of white rabbits with blue eyes** to keep in mind for your differential. Other diseases that we will not cover but may cause neurological signs include CNS neoplasias (in particular lymphoma) and the “floppy rabbit syndrome” (idiopathic acute tetraplegia with spontaneous recovery).

Encephalitozoonosis

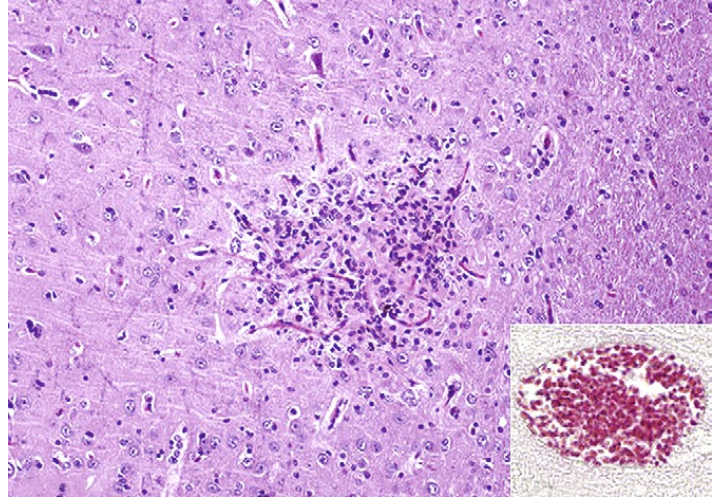
Encephalitozoon cuniculi, which literally means “brain - animal - rabbit”, is actually not an animal at all, no Ma'am! It is a **microsporidian, now classified within the kingdom Fungi**. It is primarily a pathogen of rabbits, but can also infect rodents such as guinea pigs, hamsters and mice and dogs and cats. It is a super duper common infection in rabbits with **high seroprevalence**. Not all infections lead to clinical signs.

Rabbits get infected by ingesting spores in urine-contaminated food (ew!), then it spreads everywhere specifically targetting the brain and kidneys. It also goes into the lung, heart and liver initially. It is an **intracellular obligate spore-forming organism**. *E. cuniculi* infest host cells and develop into meronts within a vacuole, and then sporonts. The vacuole may rupture and release spores into circulation, which leads to inflammation and the spores being excreted in the urine and a bit in the feces. Spores can survive outside the host for up to 6 weeks at 72F.

Lesions are mainly consistent with **multifocal non-suppurative granulomatous meningoencephalomyelitis** and meningitis, nephritis and nephrosis, cataract and pyogranulomatous uveitis.

In terms of clinical signs, as I said earlier, most infections are subclinical. Clinical signs tend to occur in older and immunocompromised rabbits. These mainly consists of **central vestibular signs** such as a head tilt, nystagmus (including **vertical nystagmus**), strabismus, circling, failling or rolling, paresis, proprioceptive deficits and other CNS signs such as paralysis, tremors, sei-

zures and incontinence. Sometimes, signs of **chronic renal disease** are also seen with the damage the organism does to the kidneys. **Myocarditis** is another common lesion.



Section of cerebrum from a rabbit with encephalitozoonosis. A granulomatous inflammatory focus is shown. Hematoxylin and eosin. The inset shows a parasitophorous vacuole at higher magnification located within the parenchyma without any associated inflammatory reaction. Acid-fast trichrome stain.



Encephalitozoonosis is also associated with a **phacoclastic uveitis** of the eye, mainly seen in younger rabbits (see below). Pyogranulomatous inflammation resulting from this rupture in the lens often protrudes into the anterior chamber, which appears clinically as a white mass coming from the iris. This is thought to be associated with *in utero* infection.

In terms of diagnostic tests, blood work is typically unremarkable except when chronic renal disease is present.



Radiographs and CT should be performed to **rule out the main other differential diagnosis: otitis media/interna**. Abdominal ultrasound may show abnormal kidneys in some bunnies. There is no definitive way to confirm the disease, but it could be highly suspected based on **serology panel including IgG and IgM**. A high IgM in particular would be interpreted as an acute infection. The magnitude of the IgG titer may also provide further evidence towards a diagnosis of *E. cuniculi*. It may be hard to interpret as the **seroprevalence is high in rabbits** in general, around 60-70%. A PCR is available, but it is considered to be unreliable as shedding in the urine mainly occurs at the beginning of the infection and spores would have to be lysed to release DNA to be detected by PCR. It has been done from CSF and extracted lens materials too. On necropsy, the lesions can be hard to see depending on the case and their severity do not correlate with the severity of the clinical disease. **Special histology stains** for the microsporidium (Gram chromotrope and others) may come handy in detecting *E. cuniculi* from brain section.

There is no good treatment for this disease. Rabbits should receive supportive care: fluid therapy, nutritional support and NSAIDs. **Benzimidazoles are commonly used to treat *E. cuniculi*, especially albendazole and fenbendazole**. Albendazole crosses the blood-brain barrier better than fenbendazole, but is associated with fatal bone marrow toxicity and other toxic effects in rabbits and is therefore used less often than fenbendazole. **Fenbendazole at 20 mg/kg PO q24h for 28 days is the recommended treatment overall. As fenbendazole can also cause bone marrow toxicity resulting in anemia, you need to monitor the CBC during treatment**. The response to therapy may be incon-

sistent or when they get better, you don't always know if it is because of the drugs or because of the vestibular system healing/compensating. Vestibular signs may be treated with midazolam 0.5-2 mg/kg IM or chronically with meclizine at 1-2 mg/kg PO q24h. Some people also use corticosteroid to decrease inflammation initially, but rabbits are a corticosensitive species so caution is advised especially when treating an infectious disease.

Last thing to consider about this disease is that it is an **uncommon zoonosis** with some reports of infections in young children and immunocompromised people.

Otitis Media and Interna

This is a very common condition of rabbits, in particular in **lop-eared rabbits**. There are 3 main pathways to getting an otitis media: **through Eustachian tubes** (typically *Pasteurella multocida*, could be other bacteria, most common in young rabbits), as an hematogenic spread of a systemic infection (rare), and most commonly from an **extension of an otitis externa leading to a mixed bacterial infection**. From an otitis media, the infection can extend to the otitis interna, and from there to the brain causing a meningitis/meningoencephalitis.

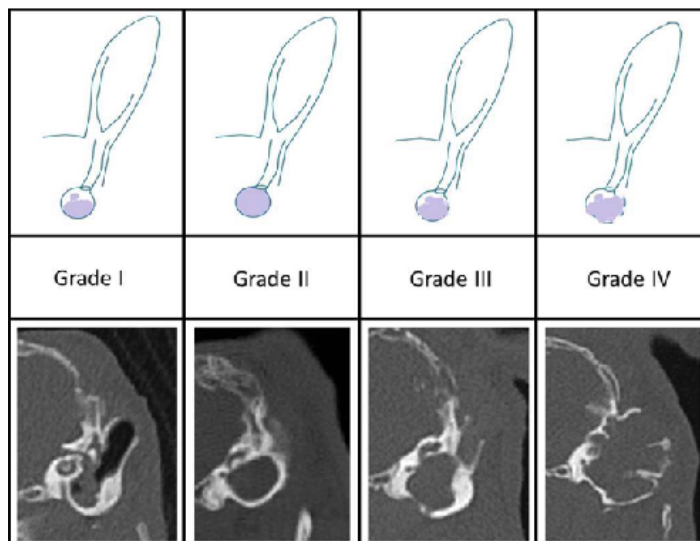
The clinical signs of otitis media include the signs related to otitis externa (pus in the external ear canal, bulging of the ear canal that can be palpated externally as a kind of mass, abnormal ear carriage), pain, and signs related to **facial nerve paralysis/paresis** (facial contracture being most common but other signs can be seen such as paresis/paralysis of the ear, eyelids, lips, and nares, a low tear production).



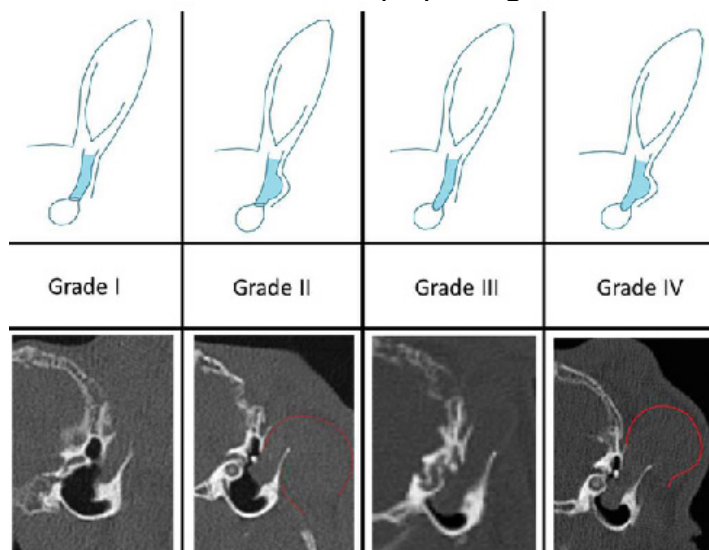
When the infection extends to the inner ear, it will

lead to a **peripheral vestibular syndrome** with signs such as head tilt, nystagmus (horizontal), strabismus, circling, falling or rolling. Rabbits are typically deaf on the affected ear because of the fluid/pus in the middle ear and alteration of the vestibulocochlear nerve (VIII).

On physical examination, examination of the ear with an otoscope may show **evidence of external otitis with thick pus in the ear and stenotic external ear canal**. Palpation of the base of the ear may reveal a thick ear canal and distended ear canal with an **ear base abscess**. On imaging, skull xrays are not very sensitive except with severe osteomyelitis of the bulla, moderate to severe thickening of the bulla or a more radiodense “filled” bulla.



CT is by far much more sensitive than xray to diagnose external, middle, and inner ear disease. CT images also allow you to grade the external and middle ear disease. See below some proposed grades:



Treatment include supportive care, as usual, antibiotics (with aerobic and anaerobic spectrum such as fluoroquinolone/metronidazole, azithromycin, chloramphenicol...), analgesics (meloxicam, tramadol, opioids) and drugs for vestibular signs such as midazolam and meclizine (see *E. cuniculi*). Ear cleansers can be used selecting some that are safe once the tympanic membrane is ruptured. Ear cleaning and flushing, including endoscopic guided ear flushing and pus debridement can be performed as well, but is hard to do and rabbits are very reactive so it has to be done under general anesthesia. Overall, **medical treatment does not work well once otitis media is there and lots of thick pus in the external ear canal**. These treatments are better done in prevention (for instance regular ear cleaning in lop-eared rabbits for instance) or with mild external otitis.

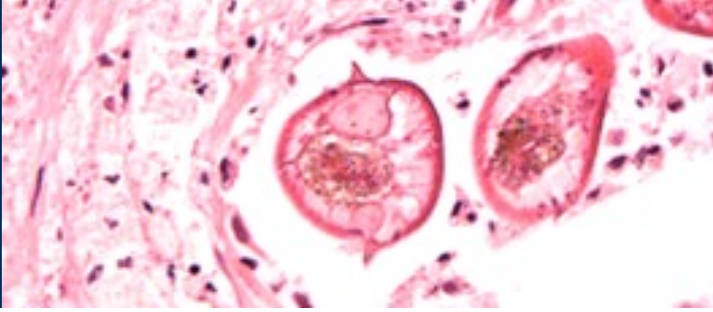
So, we often end up recommending surgery which consists of a **partial ear canal ablation and lateral bulla osteotomy (PECALBO)** to clear the pus out. At this time, a culture should be taken. See soft tissue surgery lecture for a description of the procedure.

Cerebral Larva Migrans

Baylisascaris procyonis (the raccoon roundworm) can lead to cerebral larva migrans in the rabbit. They are ingested from contaminated food or grass with raccoon feces. *Baylisascaris* eggs have an incredible capacity to survive in the environment, they are resistant to most environmental conditions and can literally survive for years. Once the eggs are ingested, the larvae are released from intestines and migrate through tissue with a predilection for the CNS. The diagnosis cannot be confirmed antemortem and is typically presumed based on a combination of clinical signs, rule out of other dis-

eases, and history (grazing outside in location where raccoons are common). There is really no treatment that works, but **oxibendazole** 60 mg/kg q24h can be tried.

On necropsy, the roundworms can be hard to find, but they have characteristic lateral alae



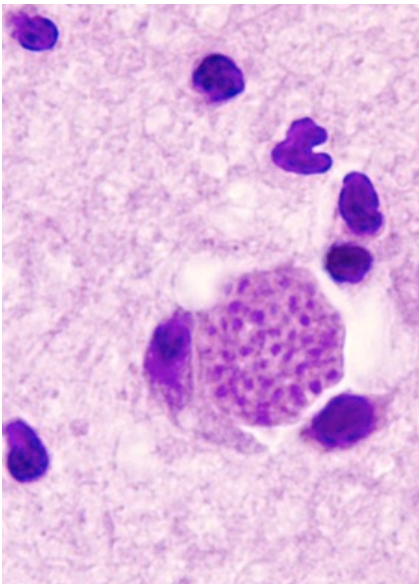
The typical clinical signs are **ataxia, tremors, vestibular signs such as head tilt and circling**.

Baylisascaris is also common in wild rodents, in particular tree squirrels.

Another parasitic disease that may lead to cerebral lava migrans in the rabbit is *Cuterebra*. It is relatively uncommon overall.

Toxoplasmosis

Toxoplasmosis is rare in rabbits but is a differential diagnosis for other parasitic encephalitis. It is obviously coming from the infamous cat (along with another parasitic disease transmissible to rabbits: cysticercosis) through fecal contamination. It tends to mainly affect immunocompromised rabbits. Diagnosis is difficult and mainly consist of serology. Lesions are similar than for encephalitozoonosis with **granulomatous meningoencephalitis**.



If toxoplasmosis is suspected, treatments that can be used include TMS, pyrimethamine, or doxycycline. Prevention is by avoiding exposure to cat feces, especially in outdoor grazing areas.

Viral Infections

There are not that many viruses that can cause neurological signs in rabbits, but some are important to know.

Herpes simplex 1 (or human herpesvirus 1) is a reverse zoonosis and rabbits get it from infected humans. It causes a nonsuppurative meningoencephalitis and can be acute and fatal. Considering how common it is in people, it seems that the transmission in rabbits is uncommon and likely requires close contact with the lesions.

The other virus that we know well is **rabies**. It is fairly rare but rabbits are definitely susceptible. Rabbits tend to have the paralytic form of the disease as well as head tremors and get the disease from exposure to wildlife.

Vertebral Fracture / Luxation / IVDD

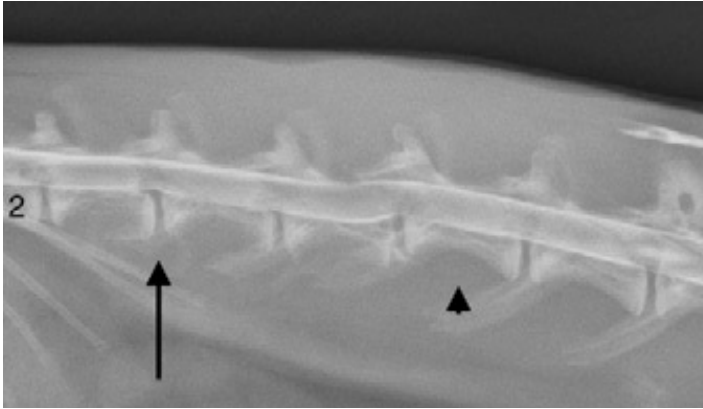
Overall, vertebral fractures are more common than luxations, which are more common than IVDD.

Rabbits are unfortunately **prone to back fractures due to their powerful back legs and relatively weak back. Improper handling and restraint, a bad jump with hyperextension of the hind quarters (especially when frightened or trying to escape) may result in vertebral fracture or luxation. The most common site of trauma is the lumbosacral region at L6-L7.**

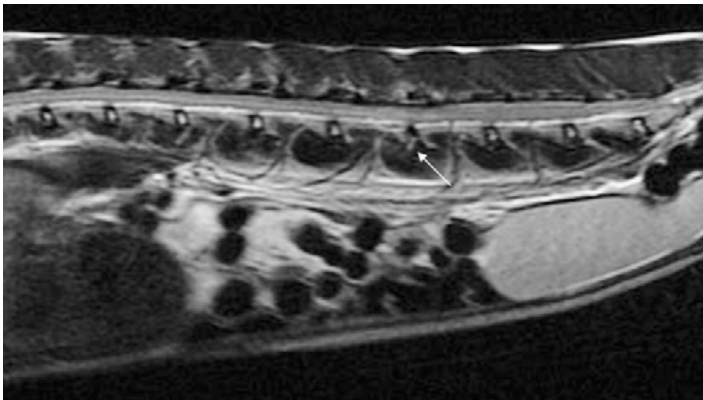
Clinical signs are those of spinal trauma and include proprioceptive deficits in the rear limbs, absence of voluntary movements, loss of superficial and deep pain perception, loss of anal tone and bladder paralysis.



The diagnosis is made by imaging: spinal radiographs, myelogram, CT, MRI.



Myelogram of a rabbit with multiple vertebral fractures and luxations. Mild dorsal protrusion of the ventral contrast column (long arrow) indicating a suspected IVD protrusion. Moderate L3-L4 subluxation with displacement of L4 dorsally (arrowhead). Mineral opacities adjacent to the L3-L4 IVD space are suspected fracture fragments of the articular facets.



Sagittal T2 MRI image of a rabbit with mild vertebral subluxation following a trauma. The arrow points to a ventral displacement of an intervertebral disc.

The treatment mainly consists of cage rest, pain management (opioids, nsoids, gabapentin, tramadol), and supportive care. If incontinent, the bladder may need to be expressed manually or may need to be catheterized. **Most of the physical recovery, if any, typically occurs during the first 6 months after the injury**, so it is hard to have a good idea of recovery and prognosis before then.

Surgical treatment for unstable vertebral fracture is possible but is tricky to accomplish as rabbits' bones are fragile and these patients may not tolerate the heavy post-operative care of such surgeries. The surgery consists of stabilizing the spine with an **external fixator and PMMA** (see 3D reconstruction from a CT-scan below).



The prognosis is overall guarded unfortunately and there are a **lot of complications** that can happen such as urinary tract infection (from atonic bladder and soiled perineum), bladder atony, urine scalding, constipation, decubital ulcers, myiasis (if goes outside) and so on.

Rabbits with mild to moderate spinal trauma might regain partial or full function, but **most fractures and luxations lead to a permanent loss of mobility**. In rabbits which do not recover completely and are not too stressed, they can be fitted to a cart (how cute is that!! just look at his face!!):



There is limited information about IVDD in rabbits.

Lead Toxicity

Lead poisoning results from the **ingestion of lead** from the household. It should be suspected in rabbits that have a tendency to chew baseboards, flooring and metallic object. Lead can be found in a variety of household items such as old paint, old linoleum and so on.



FERRET

Ferrets suffer from a variety of neurological infectious and non-infectious diseases and we will not cover all of them. For instance, we will cover the main infectious diseases, but many more may infect the neurological system in ferrets such as bacterial, fungal (cryptococcosis, blastomycosis), and parasitic organisms (toxoplasmosis, sarcocystosis). Ferrets are prone to neoplasia and **lymphoma** in particular may lead to a variety of signs depending on the affected organ system. The diagnostic approach is very similar than in dogs and cats. **Insulinoma** will lead to hypoglycemia, which in turns can cause a lot of neurological signs such as lethargy, weakness, and seizures. Please see the lecture on endocrine diseases for more info on that.

As we talked about earlier, ferrets tend to have mild **hind limb weakness with a wide range of diseases and this is not necessarily neurological**.

Viral Diseases

There are 4 viruses that can cause neurological signs in ferrets that you should know: **rabies, distemper virus, aleutian disease, and the ferret coronavirus**.

As all carnivores, ferrets are susceptible to **rabies** and should be **vaccinated annually** (ferrets are not legal in California without a permit, but dogs and cats are required to be vaccinated against rabies, so should probably other domestic carnivores such as the ferret). Clinical signs may include an ascending paralysis, ataxia, and other signs.

Ferrets are very sensitive to **canine distemper virus** and should absolutely be **vaccinated** (preferentially with a monovalent distemper vaccine). **Signs are not restricted to the CNS and include respiratory, dermatologic, and digestive signs**. For neurological signs, ferrets frequently exhibit hyperexcitability, tremor, seizures, and coma. **Mortality is close to 100%** in infected ferrets. Antemortem diagnosis is made by PCR.

Aleutian disease is caused by a **parvovirus** that has also been commonly found in minks. Typical neurological signs include **posterior ataxia and an ascending paresis/paralysis**. Diagnostic tests that can be performed are protein electrophoresis that will show a hypergammaglobulinemia, serology, and PCR.

The **ferret systemic coronavirus is a close relative to FIP (feline infectious peritonitis) and looks like its dry form**. The disease is mainly visceral causing gran-

Clinical signs include **depression, lethargy, anorexia, gastrointestinal stasis, blindness, ataxia, and seizures**.

On blood work, an **anemia** may be seen as lead interferes with heme synthesis and may lead to some hemolysis. Radiographs may show metallic foreign particles. Definitive diagnosis is achieved through blood lead concentrations obtained on a whole blood sample. Treatment includes **injectable Ca-EDTA or oral D-penicillamine**. Supportive care and treating the gastrointestinal stasis may also be required.

Fipronil Toxicity

Fipronil is a topical acaricide that is used in dogs and cats for fleas and lice (Frontline). However, do not use this product in rabbits as topical application is toxic. **Rabbits will show depression, lethargy, anorexia, ileus, ataxia, tremors and seizures (most common) from 24h to several days after application**. Rabbits may also ingest the product through grooming. Dermal absorption of topically applied fipronil in rabbits is low at 0.07% while oral absorption is higher at 30%-50%. Fipronil blocks GABA receptors in the central nervous system.

Treatment consists of **bathing** the rabbit with liquid dishwashing detergent within 48 hours of the exposure will easily remove the fipronil. **Benzodiazepines** or levetiracetam can be given to treat the seizures. **Activated charcoal** is also indicated in case oral ingestion is suspected to contribute to the toxicity. Other than that, supportive care, as usual, should be given including fluid and nutritional support.

ulomatous inflammation and masses, but **nonsuppurative meningoencephalitis** is also seen on occasion with this virus.

Intervertebral Disc Disease

IVDD is not particularly common in ferrets (as their spine is so flexible), but definitely occurs and should be a differential diagnosis for back leg issues. The most commonly affected site is L2-L3. Clinical signs are similar as in other species with spinal cord compression such as proprioceptive deficits, postural abnormalities, ataxia, decreased reflexes, and loss of superficial or deep pain sensation.

Diagnosis is made on diagnostic imaging such as radiographs (see narrowing of intervertebral space on lateral xray below), myelogram, CT, or MRI.



Treatment is as in other species. Hemilaminectomy can also be performed in ferrets to surgically decompress the spinal cord.

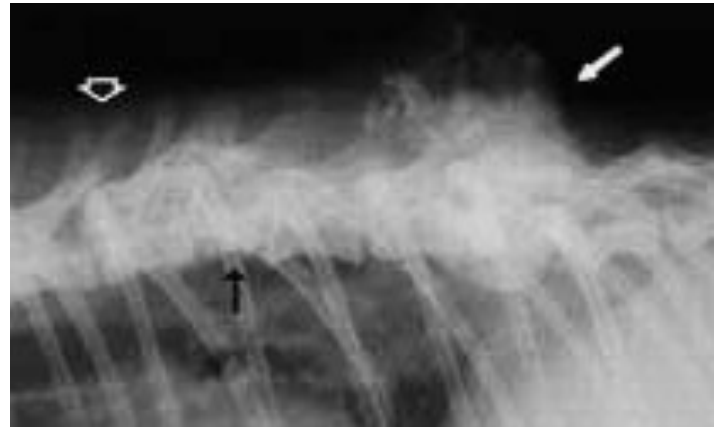
Chordoma

Chordomas are weird tumors that arise from remnants of the notochord (great, what's the notochord? it is a flexible rodlike structure of mesodermal cells that is the principal longitudinal structural element of chordates and of vertebrate embryos, it becomes part of the vertebral column). **Chordomas are typically on the tail**, which lead to minimal signs other than a mass.



However, they can less commonly be encountered in

the **cervical and thoracic vertebral column**, which will lead to spinal cord compression and ensuing neurological signs.



The tail can be amputated and some masses in the neck may be surgically removed depending on how infiltrative they are. These tumors rarely metastasize.

RODENTS

Lymphocytic Choriomeningitis

This is an **arenavirus** that can infect all types of rodents, **mainly in mice, hamsters, and guinea pigs**. Rats are resistant. It is not commonly seen in pets and is more of an issue in breeders and laboratory animals. **The normal host is the wild house mouse (*Mus musculus*), which does not show clinical signs (like lab and pet mice) and can be permanently infected.** It is transmitted through inhalation, ingestion, direct contact with body fluids and feces and also through biting insects and transplacental transmission. Clinical signs in guinea pigs and hamsters include central signs, depressed mentation, and also hind limb paresis. Diagnosis is by PCR.

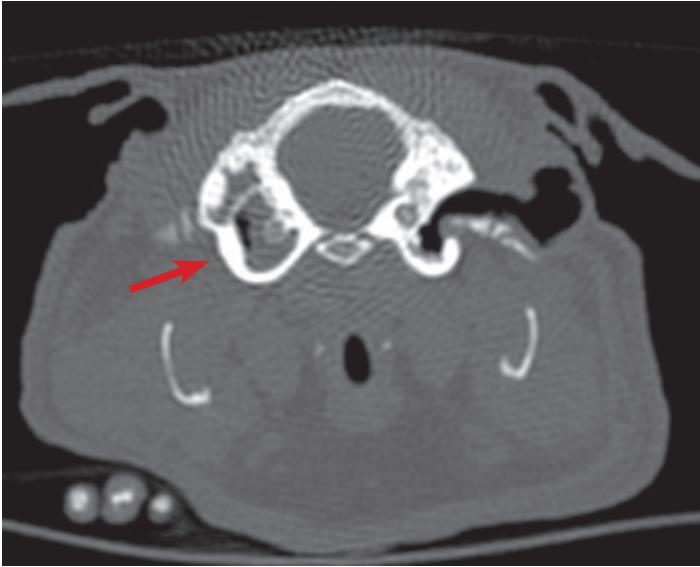
The big bummer is that it is an important **zoonosis** of rodents and can cause headache, vomiting, fever, and pregnancy issues in people. People mainly get exposed to the disease through house mice.

Otitis Media and Interna

Medial otitis, like in rabbits, is seen in **guinea pigs and chinchillas**, albeit at a lower frequency.

Within rodents, it is most commonly seen in guinea pigs. Big difference from rabbits, is that an ascending infection from the Eustachian tubes is the most common etiology. A range of bacteria may be found including ***Bordetella bronchiseptica*** (also causing respiratory

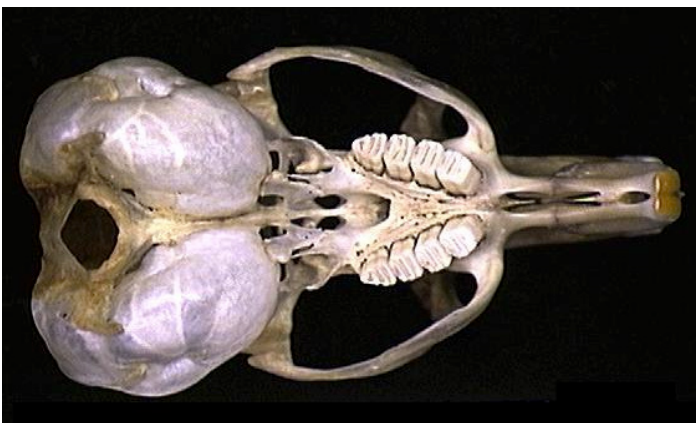
disease in these guys), *Streptococcus zooepidemicus*, *S. pneumoniae* and *pseudointermedius*. Another difference from rabbits is that the disease is **most often subclinical and found incidentally on diagnostic imaging** (see below a CT image from a subclinical guinea pig).



When clinical, **facial nerve paralysis, exposure keratitis and when progressing to otitis interna, vestibular disease** (head tilt, circling, nystagmus) are seen.

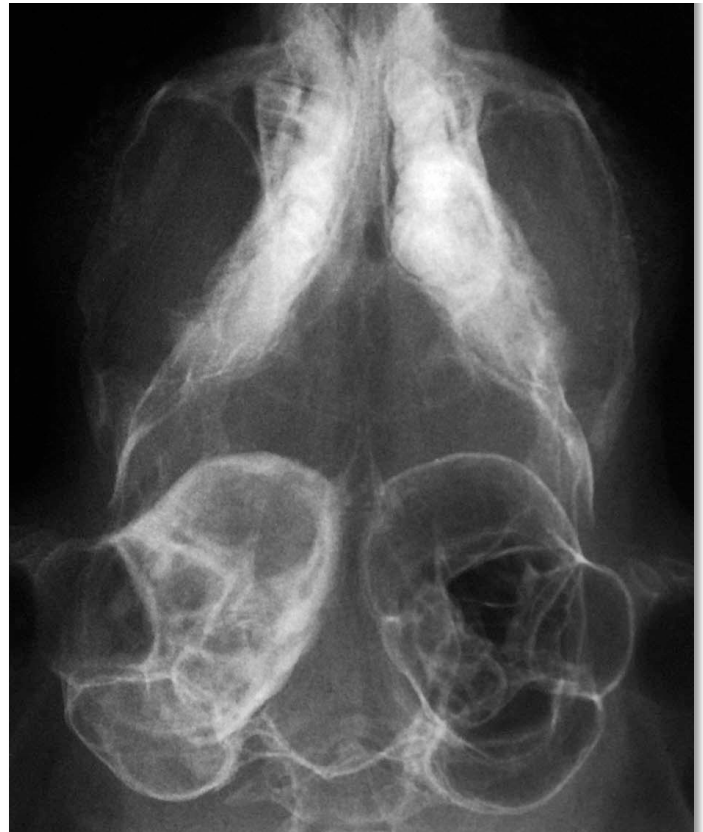
Surgical treatment like in rabbits is not as easy in guinea pigs and more complications are seen. In addition, otitis externa is not common so removing the ear canal may not be indicated. **Endoscopy-assisted myringotomy** combined with systemic antibiotics and endoscopic lavage of the bulla is an alternative treatment.

Regarding **chinchillas**, they have very large **tympanic bulla**.



The bacterial organisms recovered from the middle ear tend to be different than in guinea pigs with *Pseudomonas aeruginosa* being a common isolate.

Diagnosis is through xray or CT-scan.



Because of the size of the bulla in this species, it **can be accessed through its dorsal aspect** for obtaining culture sample, flushing and topical treatment. Fenestration into the bulla is also possible, but effectiveness is unknown as far as resolution of the otitis is concerned.



Heat Stroke

Heat stroke is **mainly seen in chinchillas** (and in rabbits a bit too). **Chinchillas tend to like cooler temperatures such as 65-80 F** and they have a thick dense fur. They will rapidly develop hyperthermia if left in

a hot humid environment, especially with low ventilation. **Severe hyperthermia can then lead to multiple organ dysfunction, endotoxemia, and DIC.** Clinical signs may include respiratory distress, tachycardia, hyperthermia, and depression. Differential diagnosis includes other causes of hyperthermia such as infection, stress, and prior seizures.

On blood work, you may see hemoconcentration or anemia (with bleeding), leukocytosis, thrombocytopenia on the CBC and azotemia, elevation of tissue enzymes, hypoglycemia, and electrolyte abnormalities on the biochemistry.

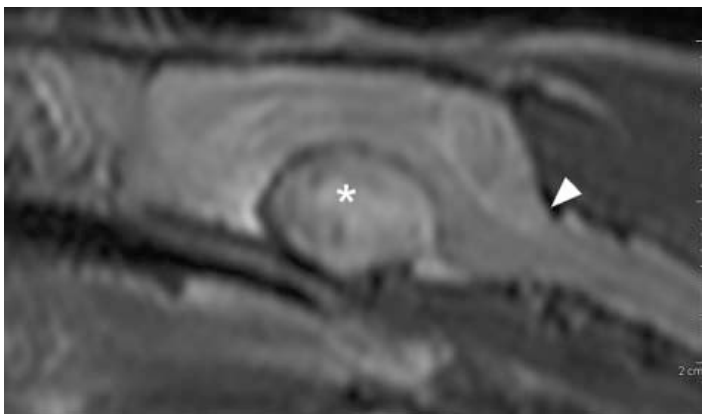
Treatment consists of **cooling the animal down, administering IV fluids**, and correcting other homeostatic imbalances (electrolytes, acid base, BP...). If the patient seems to initially make it, additional nutritional support, GI protectants, and antibiotics could be given. The **prognosis is generally guarded to poor** for those cases.

Rat Pituitary Neoplasia

This is a common tumor of pet rats, typically a **pituitary adenoma**. Some breeds of rats are particularly susceptible such as the Sprague-Dawley and Fisher strains with a reported prevalence of over 50-80% depending on studies in rats older than 2 years.

Typically clinical signs are central vestibular disease with a wide-based stance, head tilt and circling, proprioceptive deficits, and hind limb paresis and ataxia. Alteration of mentation and central blindness are also common.

The pituitary mass may be detected by advanced imaging such as contrast CT or MRI.



Midsagittal T2-weighted MRI image of the pituitary gland of a rat with a macroadenoma (asterisk).

Cabergoline, a prolactin secretion inhibitor, has

been used successfully in some cases with documentation of tumor size regression. There is some evidence that ovariectomy may prevent pituitary adenoma in Sprague-Dawley rats when done early in their life (e.g. 90 days of life).

Gerbil Epilepsy

An astonishing **20-40% of gerbils develop epilepsy** with clonic-tonic seizures so if you remember one thing about gerbil (apart from their long fluffy tail that is prone to degloving injuries), this is it (plus the tail thing). This disorder is **inherited and is caused by a deficiency in cerebral glutamine synthetase**. The disease may spontaneously resolve and gerbils may be treated with anti-seizure medications in the mean time. It is also a good spot here to be reminded that white rabbits with blue eyes and chinchillas also develop idiopathic epilepsy.

HEDGEHOGS, SUGAR GLIDERS AND OTHER RANDOM FACTS

Please **refer to the hedgehog and sugar glider lecture** for more information. Hedgehogs, beside being popular [instagram stars](#), are prone to an interesting neurological disease called the **wobbly hedgehog syndrome**. It is caused by chronic demyelination and is progressive. Hedgehogs are basically [ataxic](#) and cannot ball up and it is a progressive disease. Hedgehogs are also tumor factories (it tends to quill them).

For sugar gliders, diseases to know about are bacterial encephalitis and **nutritional disorders such as metabolic bone disease and hypoglycemia**. Their diet is kind of difficult to appropriately reproduce in captivity.

Skunks (some people like them as pets, which smells like a bad idea) can suffer from a variety of neurological diseases such as IVDD and rabies (one of the main reservoir of rabies in the wild).

Something we forgot to mention about guinea pigs as well is that they can develop seizures following mite infestation with *Trixacarus caviae*. The pruritus is so intense that it goes beyond the epileptic threshold.

The Ten Most Important Facts

1. The neurological examination is very similar in small exotic mammals than in domestic carnivores. The main differences pertain to the interpretation of certain postural reactions and reflexes such as the absence of menace response in most rabbits and rodents and inconsistencies of the perineal reflex and hindlimb proprioceptive placement.

2. *Encephalitozoon cuniculi* is a microsporidian organism which infects the CNS, heart, and kidneys of rabbits and leads to a nonsuppurative meningoencephalitis. The most common clinical signs include a central vestibular syndrome and phacoclastic uveitis (in the young). The seroprevalence is high, but we still use a serology panel (IgG and IgM) for diagnosis. Fenbendazole is the treatment. It is a **zoonosis**.

3. Otitis media and interna is seen frequently in rabbits and not uncommonly in guinea pigs and chinchillas. In rabbits, it is mainly associated with lop-eared rabbits and otitis externa whereas in rodents it is mainly caused by an ascending infection through the Eustachian tubes. Treatment is mainly surgical in rabbits consisting of a PECALBO and antibiotics. CT is the most sensitive imaging method for this.

4. Rabbits are prone to lumbar vertebral fractures and luxations. Prognosis is typically guarded to poor, if improvement is seen, it tends to occur mainly over the following 6 months.

5. *Baylisascaris procyonis*, the racoon worm, is an important neurologic disease of rabbits grazing outside and wild rodents and is a diagnosis of exclusion.

6. Fipronil is toxic in rabbits.

7. Ferrets have 4 main viral diseases causing neurological signs: rabies, distemper virus, aleutian disease, and the ferret systemic coronavirus (like the dry form of FIP in cats).

8. Lymphocytic choriomeningitis is an important **zoonosis** of rodents that is carried asymptotically by the house mouse and can infect guinea pigs and hamsters.

9. Pituitary adenoma is a common central tumor of rats that will lead to central vestibular signs, ataxia, and central blindness. Carbergoline has been used as a treatment.

10. Lots of gerbils seizure. It may resolve spontaneously over time. Also, the wobbly hedgehog syndrome.